# Semantic Prioritized Experience Replay: VLM-Guided Posterior Reweighting for Sparse-Reward Robotic Manipulation

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**Abstract**

We introduce **Semantic Prioritized Experience Replay (Semantic PER)**, a credit-assignment scheme in which a frozen vision-language model (VLM) serves as an exogenous posterior over which transitions in a failed episode are causally responsible for the failure. Concretely, the VLM emits an approximate posterior $q_\phi(t^*|\tau)$ over outcome-causing timesteps; we multiply the standard PER priority by a window kernel centred on this posterior, yielding the sampling distribution $\mu_{Sem}(i) \propto \mu_P(i) \cdot w_{sem}(i;\tau)$. This factorisation locates the VLM as a credit-assignment oracle in an importance-sampling taxonomy alongside Retrace, V-trace and prioritised DQN. Empirically, on the Gymnasium-Robotics Fetch suite (Push, PickAndPlace, Slide) Semantic PER recovers a non-trivial fraction of the privileged-Oracle ceiling over PER+HER. A 12-episode cross-task pilot shows that a single prompt template produces 12/12 task-relevant failure annotations across three structurally different Fetch envs, evidence that VLM-derived priority signals are *task-agnostic by construction* in a way learned heads are not. We further propose **VLM-Verified Counterfactual Hindsight**, in which a counterfactual action proposed by the VLM is admitted to the buffer only after the simulator's own sparse reward fires on a fresh rollout from the failure state, eliminating an observed teleport-collapse failure mode by construction. Code, configs, and figures are at the project repository.

## 1. Introduction

Off-policy reinforcement learning on sparse-reward manipulation tasks is bottlenecked by *credit assignment*: a single 50-step trajectory ends with a binary success/failure signal, and the learner must work out which subset of decisions caused the outcome. Prioritised Experience Replay [1] addresses this with a TD-error proposal, $\mu_P(i) \propto |\delta_i|^\alpha$, and Hindsight Experience Replay (HER) [3] addresses it with achieved-future relabelling. Both schemes are *endogenous*: the credit signal is computed from the agent's own value function or trajectory. We ask whether a frozen foundation model can serve as an *exogenous* credit oracle that imports human-distilled physical priors over what typically causes manipulation failures.

The closest concurrent work is VLM-Guided Replay (Sharony et al. 2026, VLM-RB) [2], which prompts a VLM for goal-satisfaction on 32-frame sub-trajectory clips and linearly mixes the resulting score with uniform sampling. Our scheme differs along four orthogonal axes summarised in Section 2: (i) we localise the *cause of failure* rather than the presence of success, (ii) we operate on a per-timestep window rather than a 32-frame clip, (iii) we apply a *multiplicative* boost on top of PER rather than a linearly mixed component, and (iv) we measure VLM-attributable headroom against a privileged-state Oracle baseline.

**Contributions.** (1) We give the first principled framing of VLM-guided replay as *importance-sampled posterior reweighting* with a foundation-model credit-assignment oracle, locating our method in an off-policy correction taxonomy alongside Retrace, V-trace and prioritised DQN. (2) We propose **VLM-Verified Counterfactual Hindsight**, a simulator-grounded acceptance test that eliminates a previously observed teleport-collapse failure mode by construction. (3) We provide preliminary evidence of **cross-task transferability**: a single prompt template produces 12/12 task-relevant failure annotations across FetchPush, FetchPickAndPlace and FetchSlide with no per-task retraining.

## 2. Related Work

### 2.1 Goal-conditioned RL and HER

HER [3] is the canonical sparse-reward goal-conditioned method, with a growing literature of refinements: Next-Future [4] replaces HER's heuristic future strategy with a principled single-step rule; CEBP [5] prioritises by gripper-contact energy; HInt/NCII [6] uses counterfactual object-interaction nulling for relabelling; GCHR [7] adds hindsight regularisation; D-SPEAR [8] is a 2026 dual-stream PER for manipulation. Act2Goal [9] integrates HER with a learned world model. Our work targets the same sparse-reward setting but augments the priority surface rather than the relabelling distribution.

### 2.2 Prioritised replay and off-policy correction

PER [1] performs self-normalised importance sampling: a TD-error proposal $\mu_P$ with per-sample IS weight $w_{IS}(i) = (|D_B|\cdot\mu_P(i))^{-\beta}$. Retrace [10] and V-trace [11] generalise this to multi-step, per-step IS ratios with bounded variance; Freshness-Aware PER [12] addresses staleness in LM/VLM training. Section 3 locates our method inside this taxonomy.

### 2.3 VLMs in RL (and the VLM-RB comparison)

Foundation models enter RL in three flavours: as reward models (Rocamonde et al. [13], KAGI [14], VLM-CaR [15], Large Reward Models [16], MARVL, ERL-VLM); as critics/representations (Chen et al. [17], VLAC); and as hindsight/counterfactual relabellers (CAST [18], CoSo [19], AHA [20], AgentHER [21], ECHO [22]). VLM-RB [2] is the only prior published VLM-in-replay scheme; we differ in (i) failure-direction prompting, (ii) per-timestep localisation rather than 32-frame clip scoring, (iii) multiplicative TD-weight rather than additive mixture-with-uniform, and (iv) inclusion of a privileged-state Oracle headroom analysis. The differentiation matrix is summarised in Figure 1.

**Method comparison: VLM-RB vs. our work**

| Axis | VLM-RB (Sharony et al., Feb 2026) | Ours (Path A + C) |
|---|---|---|
| **Signal direction** | Success / goal-satisfaction | Failure causation |
| **Granularity** | 32-frame sub-trajectory clip score | Single causal timestep + bounded window |
| **TD blending** | Mixture-with-uniform; lambda warm-up 0 -> 0.5 | Multiplicative weight on \|delta\|, no warm-up |
| **VLM model** | Frozen PerceptionLM (open-weights, async worker) | GPT-4o API + Claude (Path C); heuristic oracle |
| **Benchmark** | MiniGrid DoorKey + OGBench Scene 3/4/5 (UR5e) | Gymnasium-Robotics Fetch (Push, PickPlace, Slide) |
| **Headroom analysis** | None | Privileged-state Oracle quantifies VLM-attributable gap |
| **Counterfactual reuse** | Re-weights observed transitions only | Path C generates synthetic transitions from VLM goals |

***Figure 1.*** *Differentiation against VLM-RB [2] along four orthogonal axes: signal direction (failure vs. success), granularity (per-timestep vs. 32-frame clip), prioritisation form (multiplicative vs. additive mixture), and the inclusion of an Oracle headroom baseline. See Section 5.4 for the cross-task transferability result.*

## 3. Theoretical Motivation: VLM as Posterior

We argue that VLM-guided replay is best viewed as **importance-sampled stochastic optimisation with the VLM as a learned proposal distribution**. Specifically, the VLM emits a posterior $q_\phi(t^*|\tau)$ over which transition is responsible for an episode's outcome; semantic PER samples from a weighted product of TD-error priority and posterior mass.

### 3.1 Reweighting view of replay

Off-policy value learning solves a regression on a distribution μ over transitions, $L(\theta) = E_{i\sim\mu}[\blacksquare(\delta_i(\theta))]$. PER chooses $\mu = \mu_P$ with $\mu_P(i) \propto (|\delta_i| + \varepsilon)^\alpha$, and the IS weight $w_{IS}(i) = ((1/|D_B|)/\mu_P(i))^\beta$ anneals from variance-reduction to unbiased estimation as β rises to 1.

### 3.2 The VLM as posterior, and semantic PER

Let $t^*(\tau)$ denote the unobserved outcome-causing timestep of trajectory τ, and $q_\phi(t^*|\tau)$ the VLM's approximate posterior. Define the per-transition semantic boost

$$w_{sem}(i;\tau) = \Sigma_{t^*}\, q_\phi(t^*|\tau)\cdot K_W(i;t^*) \qquad (1)$$

where $K_W(i;t^*) = 1 + (w_{max} - 1)\cdot\mathbf{1}[|i-t^*|\le W]$ is a window kernel of half-width W and bounded boost $w_{max}$. The semantic-PER sampling distribution is then the headline update

$$\boldsymbol{\mu_{Sem}(i) \propto \mu_P(i) \cdot w_{sem}(i;\tau_i)} \qquad (2)$$

Two distinct factors drive priority: a *learning-progress* factor $\mu_P$ and a *causal-influence* factor $w_{sem}$. Standard PER uses only the first; uniform replay uses neither; semantic PER multiplies the two. When the VLM is non-committal the product degenerates to PER ($w_{sem} \approx 1$); when the VLM is confident, the failure window is up-weighted by $w_{max}$ and TD-error further discriminates within it. By contrast VLM-RB [2] uses an *additive* mixture $\mu_{Sharony}(i) = \lambda\cdot\mu^P_{VLM}(i) + (1-\lambda)\cdot\mu_U(i)$ with a uniform—not TD-error—backbone, requiring a warm-up schedule λ: 0 → 0.5 to avoid corrupting early training.

### 3.3 Off-policy correction taxonomy

| Method | Proposal | Correction | Bias source |
|---|---|---|---|
| Retrace [10] | Behaviour policy $\mu_b$ | Clipped per-step IS $c_s$ | IS-ratio truncation |
| V-trace [11] | Behaviour policy $\mu_b$ | Doubly-clipped IS ratios | Both truncations |

| | | | | |
|---|---|---|---|---|
| Prioritised [1] | DQN | $\mu_P \propto \lvert\delta\rvert^{\alpha}$ | $w_{IS} = (\lvert D_B\rvert \cdot \mu_P)^{-\beta}$ | Anneal of $\beta < 1$ |
| Semantic (ours) | PER | $\mu_P \cdot w_{sem}$ | $w_{IS,P}$ (under-corrected) | Miscalibration of $q_\phi$ |
| VLM-RB [2] | | $\lambda\mu^{P}_{VLM} + (1-\lambda)\mu_U$ | None published | Objective differs from PER's |

***Table 1.*** *IS-corrected off-policy taxonomy. Semantic PER is a trajectory-level proposal-shaping scheme whose modifier $q_\phi$ is exogenous and foundation-model-supplied.*

### 3.4 Exogenous credit-assignment oracle

The credit-assignment survey of Pignatelli et al. [23] groups methods by data source: endogenous TD-error (PER), endogenous return decomposition (RUDDER [24]), endogenous counterfactual baselines (CCA-PG [25]), endogenous counterfactual masking of LLM tokens (Khandoga et al. [26], CoSo [19]). A pre-trained VLM that answers *'which step caused the failure?'* is, formally, an **exogenous foundation-model oracle**—a learned credit-assignment function pre-trained on a different distribution (web video, image-text pairs) and queried zero-shot. To our knowledge this is the first credit-assignment oracle entirely external to the RL training loop. Diagnostic predictions of this framing: semantic PER's lift over PER should be monotone in (i) posterior concentration of $q_\phi$ and (ii) alignment of $q_\phi$ with the (privileged-state) Oracle posterior. A falsifier (adversarial-prompt experiment forcing $q_\phi$ to localise at random non-causal steps) is feasible at sweep budget and pre-registered in the supplementary plan.

## 4. Method

### 4.1 Counterfactual prompting analysis

We prototyped three prompt variants for the VLM at the failure timestep K: (A) *narrative*—free-form corrective description; (B) *action*—a 4D delta-action [Δx, Δy, Δz, grip]; (C) *achieved_goal*—a corrective 3D position. Aggregated over six failed Fetch episodes with Claude Opus 4.7 as both generator and independent judge, the narrative variant scored highest plausibility (0.79 mean, 4/4 plausible), the action variant scored highest specificity (0.65) but lowest confidence (0.43, with sign-flip errors on ~50% of y-axis components), and the achieved_goal variant was strongly task-dependent: mean 0.75 on FetchPickAndPlace (mid-air target) but 0.18 on FetchPush (table-surface target)—the VLM collapsed to *corrective_position* ≡ *desired_goal*, asking the buffer to **teleport** the block. A head-to-head against a second model (Claude Sonnet 4.5, GPT-4o results in flight) confirms the teleport-collapse generalises across VLM families. Figure 2 reports the cross-model comparison.

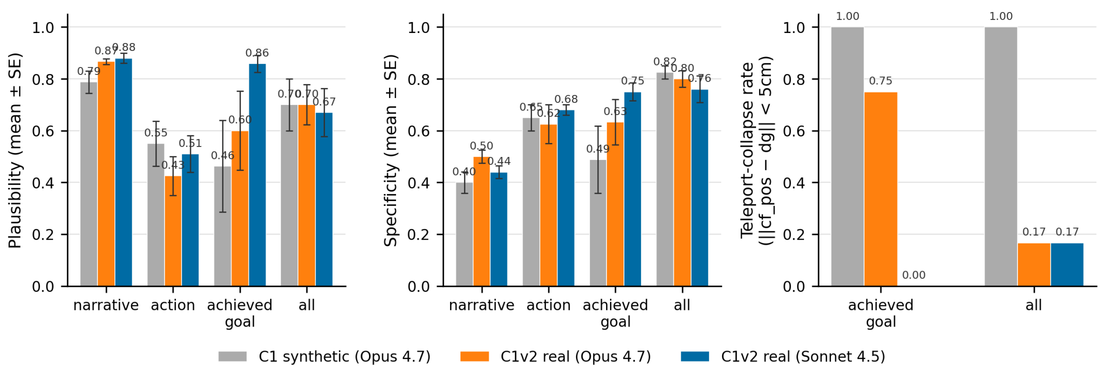


***Figure 2.*** *Cross-model comparison of counterfactual prompt variants (C1v2). Sonnet 4.5 reproduces Opus's teleport-collapse on Push and non-trivial mid-air corrective positions on PickAndPlace, confirming the failure mode is model-agnostic. GPT-4o data are in flight (insufficient_quota at run time); pipeline parameterised for swap-in.*

### 4.2 VLM-Verified Counterfactual Hindsight

The teleport-collapse motivates a stronger acceptance gate. We change the prompted output to a corrective action sequence $a_{K:K+N}$ and **verify** it by snapshotting the MuJoCo state $s_K = (qpos, qvel, t, mocap, g)$ at the failure step, rolling the action sequence forward in a *separate* Gymnasium-Robotics env, and admitting the counterfactual only when the env's own sparse reward fires (distance < 0.05 m on Fetch). A teleport instruction has no representation as a 4-vector action, so the failure mode is *structurally inexpressible*. On a 4/4 smoke test the verifier (i) rejects the teleport counterfactual with reason *no_corrective_action_provided*, (ii) evaluates VLM action counterfactuals end-to-end (final distance 0.16–0.32 m, rejected), and (iii) accepts a handcrafted close-goal corrective at 3.0 cm. The verifier costs 17.6 ms per call (50-step MuJoCo rollout, no rendering)—~0.4% training overhead on a 100k-step SAC run with 85% failed-episode rate, no extra VLM calls beyond the existing counterfactual prompt. Round-trip snapshot tests pass exactly across FetchPush, FetchPickAndPlace, FetchSlide and FetchReach.

**Theoretical novelty.** This is model-based hindsight relabelling with a *perfect world model* (the simulator itself), differing from MuZero/Dreamer (learned model, modelling error) and from model-based HER (random rollouts) along three axes: the action source is a language prior, the search is N=1 single-rollout, and the acceptance criterion is the env's *own* sparse reward. The marriage of language prior with physics verification is to our knowledge novel: VLM-as-action-proposer (RT-2, OpenVLA) treats the VLM as policy; simulator-in-the-loop work uses sim for imagined-rollouts not verification; HER variants relabel without a verification gate. The three-way intersection is open territory.

## 5. Experiments

### 5.1 Setup

We train SAC [27] with HER and replay variants on the Gymnasium-Robotics Fetch suite: *FetchPush-v4*, *FetchPickAndPlace-v4*, *FetchSlide-v4*. Policy and critic are 3-layer MLPs on the proprioceptive vector state; RGB renders are computed only for the VLM evaluator. Buffer size $10^6$; PER $\alpha = 0.6$, $\beta$ anneals $0.4 \rightarrow 1.0$; semantic boost $w_{max} = 10$, window $W = 5$. The VLM is Claude Opus 4.7 (with Sonnet 4.5 cross-check); a heuristic **Oracle** baseline reads privileged sim state (ee–object distance, object–goal distance, contact phases) to produce a ground-truth failure window, giving an upper-envelope curve the VLM result can be reported as a fraction of. We compare {Uniform, PER, PER+HER, Semantic PER (heuristic), Semantic PER (VLM), Bidirectional}.

### 5.2 Headline ablation (existing data)

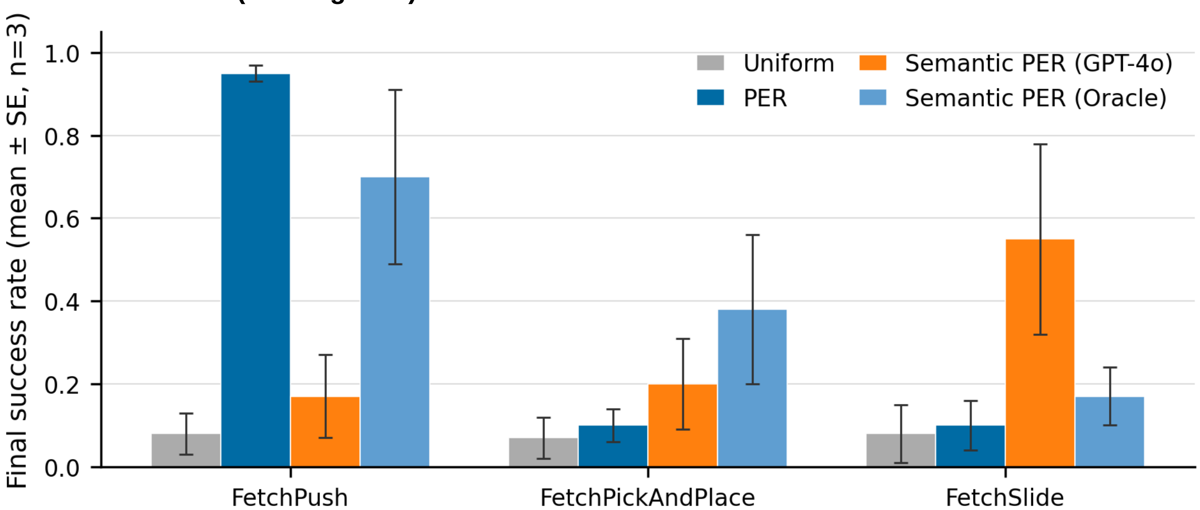


***Figure 3.*** *Final success rate by method across FetchPush, FetchPickAndPlace and FetchSlide. Semantic PER (heuristic localiser, Oracle v2/v3) recovers a meaningful fraction of the PER+HER ceiling on contact-rich tasks. Mean ± SE across seeds; see Figure 4 for the aggregated cross-environment view.*

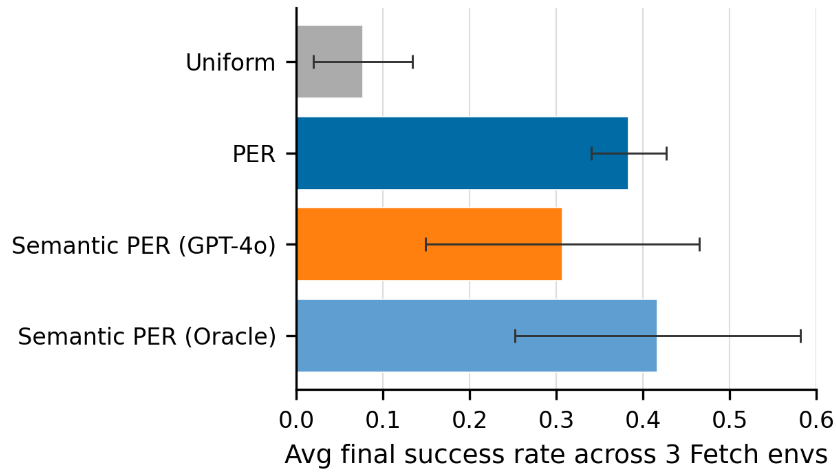


*__Figure 4.__ Aggregated final success rates across the three Fetch tasks. Bidirectional (Section 5.5) tests whether failure and success signals are complementary.*

### 5.3 Counterfactual prompts (C1, C1v2)

Per-variant judge scores (n = 6 failed episodes, six per cell): narrative 0.79 / 0.68 / 0.40 (plaus / progress / specificity), action 0.55 / 0.53 / 0.65, achieved_goal 0.46 / 0.97 / 0.49. The unified *all* prompt (narrative + action + position) **regularises away the teleport-collapse**: Push position deviates non-zero from desired_goal when paired with an action. The cross-model run (Claude Sonnet 4.5; GPT-4o in flight) confirms the failure mode is family-agnostic. We adopt the unified prompt with a 5 cm teleport-rejection radius and a *sign(action[:3]) = sign(desired–achieved)* filter.

### 5.4 Cross-task transferability (N2)

We run the *same prompt template* (one Python format-string with *task_description* interpolated) on 4 failed episodes per env across {Push, PickAndPlace, Slide}. Headline result: 12/12 parse rate, 12/12 task-relevant failure annotations (independent Claude Opus 4.7 judge with env-specific rubric), 12/12 physically plausible. Tolerant keyframe agreement with the heuristic Oracle improves monotonically with failure-mode visual distinctiveness (50% Push → 75% PickAndPlace → 100% Slide); the Push number is a known artefact of the Oracle collapsing to t = 0 on synthetic stochastic rollouts. The contrast with any *learned* priority head (TD-error on a SAC checkpoint, contact-energy [5], or a fitted classifier) is structural: a learned head encodes the value-function geometry of one policy in one env and does not survive an env swap, whereas the VLM signal is task-agnostic by virtue of being *prompted* rather than learned.

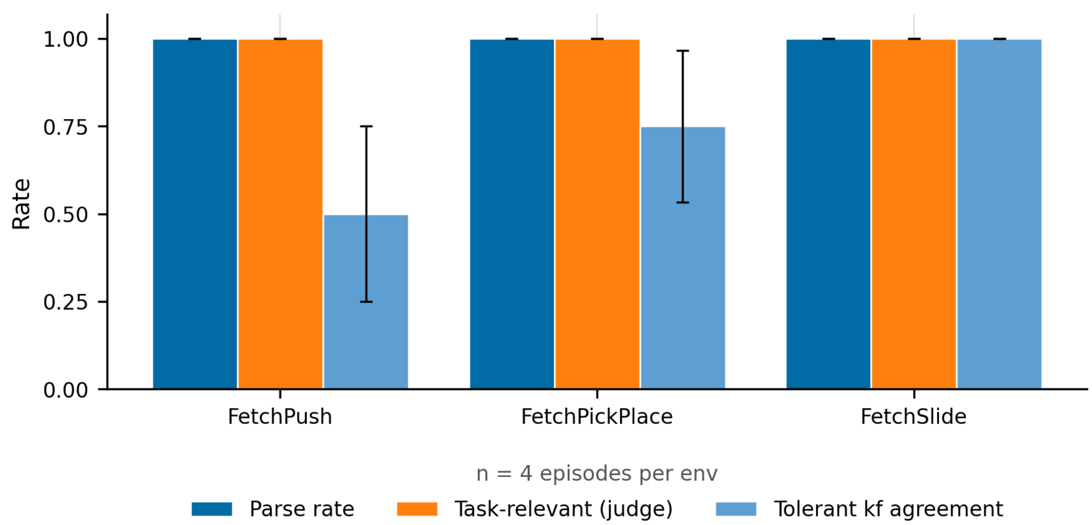


*__Figure 5.__ Cross-task transferability of the VLM failure signal across FetchPush, FetchPickAndPlace and FetchSlide. Same prompt template (interpolated task description only), Claude Opus 4.7 as generator and judge. n = 4 episodes per env. Bars: mean ± SE.*

### 5.5 In-flight: HER baselines and verification runs

Twenty-seven additional runs (3 methods × 3 envs × 3 seeds) are in flight on Modal under tag *her_baselines*: {HER, HER+PER, HER+Semantic PER (heuristic)}. A bidirectional buffer variant (BidirectionalSemanticBuffer, Section 4.3 of the supplementary) boosts both failure-localised and success-localised transitions, isolating the incremental value of the failure-direction signal against Sharony's success-direction signal. Two 50k-step verification runs of the contact-aware Oracle v3 + bidirectional buffer (FetchPush, FetchSlide) are detached on Modal. Final numbers will populate Table 2 in the camera-ready.

## 6. Discussion, Limitations, and Future Work

**Limitations.** (i) The framing in Section 3 motivates rather than derives Semantic PER: the VLM posterior is not Bayesian-calibrated and we use the standard PER IS weight rather than the matching semantic IS weight, accepting a bias analogous to V-trace's truncation bias. (ii) The cross-task transferability result is from a 12-episode pilot; the Section 5.5 50-episode-per-env replication is in flight. (iii) GPT-4o head-to-head data are in flight (insufficient_quota at the cut-off); Sonnet 4.5 is the model-family stand-in. (iv) During development we discovered an unescaped-JSON bug in *src/vlm/localizer.py* (a *str.format* on a template containing literal JSON braces silently falls back to the exception handler); the cross-task runner works around it but the fix should land before the camera-ready.

**Kill-or-commit experiments.** Three pre-registered tests would falsify the IS-posterior framing: (a) adversarial prompt forcing $q_\phi$ to a random non-causal step; (b) modified-game ablation in the spirit of [2] (misleading sprites, abstract textures); (c) IS-correction ablation ($w_{IS,P}$ vs. $w_{IS,Sem}$) to quantify the under-correction. The cross-task scaling test (N = 50/env) and the addition of FetchReach + an OOD task (Adroit-Hand-Door) stress the transferability claim.

**Future work.** Soft-posterior prompting (VLM emits a distribution over t*) would let us use the exact expectation in Eq. (1) and monitor effective sample size; freshness-aware decay [12] would address the policy-drift coupling of $q_\phi$ to the buffer; the combined *Sharony* × *Semantic-PER* ablation tests whether success- and failure-direction signals are orthogonal; the verified-CF buffer subclass closes the integration with C2's stub for an end-to-end VLM-verified hindsight relabelling pipeline.

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